1. Run report: run-20261017T0201Z

Study `research/generic_dlp_memory_parallelism_20261017` · **protocol** `PROTOCOL.md` (frozen before timing; amendment 1) · **Conductor task** T-1 · **PR** 2048

This report describes what ran and what was recorded. The study's `README.md` and `REPORT.pdf` carry the interpretation.

1.1. What ran

- **Baseline A and A′** (the A/A control): `main` at `b1bf5b8935f68257dfdd3fc542cb2fef41db741c`.
- **Candidate B**: `c699775dde4641313c5d045d6075497604db9657` on `claude/generic-dlp-memory-parallelism`.
- **Build**: cargo build --release (default profile, no target-cpu flags), rustc 1.97.0 (2d8144b78 2026-07-07).
- **Host**: Intel(R) Xeon(R) Processor @ 2.10GHz, 4 logical CPUs, KVM guest.
  - Features: adx avx2 avx512f avx512ifma bmi2 pclmulqdq popcnt .
  - Caches: L1d cache: 192 KiB (4 instances);L2 cache: 8 MiB (4 instances);L3 cache: 260 MiB (1 instance);.
  - Kernel 6.18.44-fc-v139 transparent huge pages `always [madvise] never`.
- **Isolation** (as written at run start): tools/isolated_bench.py run --cpus 3 (1-thread arms) or --cpus 0-3 (4-thread arms); records in raw/isolation.jsonl.
  - **Amendment**: Amendment 1 (PROTOCOL.md, 2026-10-17, before any multi-thread arm ran): the isolation and commands fields above were written at run start; the multi-thread arms ran as parts bsgs3 and rho3 with 3 threads on --cpus 1-3, because tools/isolated_bench.py refuses to reserve every CPU of the 4-vCPU guest.
  - Isolation level earned: shared KVM guest, so L0 by ecbench's scale.
  - Contended runs are flagged per record and left out of every statistic.
- **External tool**: PARI/GP [2, 15, 4] (Ubuntu noble pari-gp), single-threaded, elllog timed by getabstime().
- **Commands**: `research/generic_dlp_memory_parallelism_20261017/scripts/run_arms.sh <raw> <baseline_bin_dir> <candidate_bin_dir> {bsgs1,bsgs4,rho1,rho4}`
  `research/generic_dlp_memory_parallelism_20261017/scripts/pari_elllog.sh <raw> <role> <bits> <k> <label>`
- **Runs**:
  - BSGS: 1 thread at 40, 44, 48 and 52 bits; 3 threads at 48 bits. Kernels `chain` and `paired` (`K = 16`).
  - Rho: 1 and 3 threads at 32–52 bits.
  - 7 rounds (5 at 52 bits), arms rotated `A B A′ / B A′ A / A′ A B`.
  - One target per process.
  - PARI/GP: 3 repetitions per curve.
- **Interruptions** (`interruptions.txt`), none of which reran or overwrote a recorded sample:
  1. A lock conflict stopped the script, which then resumed.
  2. The live output was moved out of the repository between samples.
  3. The multi-thread parts started under amendment 1: 3 threads on CPUs 1–3, because the isolation tool refuses to reserve every CPU.

1.2. Recorded values

Times are in seconds, from each binary's own timer. **A/B** is the ratio of the medians of A and B. **Per-round** gives the range of the per-round paired ratios. The **verdict** applies the protocol's rule: "improves: every per-round A/B ratio above the largest per-round A/A2 ratio (and above 1); regresses: every one below the smallest (and below 1); otherwise no measurable change".

part	bits	kernel	timer	A med.	B med.	A/B	A/B per round	A/A′ per round	verdict	pinned
bsgs1	40	chain	build_s	0.0693	0.0514	1.348	[0.972, 1.554]	[0.771, 1.469]	no measurable change	identical
bsgs1	40	chain	search_s	0.0384	0.0332	1.157	[0.939, 1.431]	[0.857, 1.502]	no measurable change	identical
bsgs1	40	chain	total_s	0.109	0.0858	1.27	[0.959, 1.492]	[0.842, 1.481]	no measurable change	identical
bsgs1	40	paired	build_s	0.0346	0.0249	1.39	[1.021, 1.559]	[0.852, 1.075]	no measurable change	identical
bsgs1	40	paired	search_s	0.0227	0.0169	1.34	[1.034, 1.513]	[0.788, 1.057]	no measurable change	identical
bsgs1	40	paired	total_s	0.0565	0.0417	1.354	[1.026, 1.54]	[0.825, 1.067]	no measurable change	identical
bsgs1	44	chain	build_s	0.3339	0.2315	1.442	[1.287, 1.616]	[1.025, 1.229]	improves	identical
bsgs1	44	chain	search_s	0.0142	0.0104	1.372	[1.196, 1.632]	[1.008, 1.294]	no measurable change	identical
bsgs1	44	chain	total_s	0.3482	0.2415	1.442	[1.287, 1.617]	[1.024, 1.222]	improves	identical
bsgs1	44	paired	build_s	0.2176	0.1087	2.003	[1.318, 2.678]	[0.913, 1.112]	improves	identical
bsgs1	44	paired	search_s	0.0098	0.0052	1.885	[1.388, 2.13]	[0.703, 1.107]	improves	identical
bsgs1	44	paired	total_s	0.2265	0.1142	1.984	[1.32, 2.652]	[0.902, 1.108]	improves	identical
bsgs1	48	chain	build_s	1.4303	1.0971	1.304	[1.242, 1.401]	[0.932, 0.996]	improves	identical
bsgs1	48	chain	search_s	0.0048	0.0035	1.391	[1.353, 1.5]	[0.885, 1.021]	improves	identical
bsgs1	48	chain	total_s	1.4351	1.1003	1.304	[1.243, 1.401]	[0.932, 0.996]	improves	identical
bsgs1	48	paired	build_s	1.0384	0.5522	1.88	[1.737, 1.957]	[0.957, 1.028]	improves	identical
bsgs1	48	paired	search_s	0.004	0.002	2	[1.85, 3.294]	[0.537, 1.436]	improves	identical
bsgs1	48	paired	total_s	1.0429	0.5543	1.881	[1.739, 1.957]	[0.956, 1.028]	improves	identical
bsgs1	52	chain	build_s	6.5145	4.9359	1.32	[1.24, 1.402]	[0.999, 1.015]	improves	identical
bsgs1	52	chain	search_s	0.0019	0.0014	1.357	[1.286, 1.769]	[0.613, 1.353]	no measurable change	identical
bsgs1	52	chain	total_s	6.5163	4.9373	1.32	[1.241, 1.402]	[0.999, 1.015]	improves	identical
bsgs1	52	paired	build_s	4.6099	2.6453	1.743	[1.667, 1.847]	[0.947, 1.038]	improves	identical
bsgs1	52	paired	search_s	0.0015	0.0009	1.667	[1.5, 1.889]	[0.875, 1.133]	improves	identical
bsgs1	52	paired	total_s	4.6114	2.6462	1.743	[1.667, 1.847]	[0.947, 1.038]	improves	identical
bsgs3	48	chain	build_s	0.574	0.4303	1.334	[1.234, 1.447]	[0.94, 1.167]	improves	identical

bsgs3	48	chain	search_s	0.0061	0.0038	1.592	[0.912, 2.286]	[0.627, 1.346]	no measurable change	identical
bsgs3	48	chain	total_s	0.581	0.435	1.336	[1.232, 1.44]	[0.942, 1.165]	improves	identical
bsgs3	48	paired	build_s	0.4107	0.2744	1.497	[1.385, 1.845]	[0.853, 1.12]	improves	identical
bsgs3	48	paired	search_s	0.0044	0.0025	1.76	[1.538, 2.857]	[0.848, 1.304]	improves	identical
bsgs3	48	paired	total_s	0.4152	0.2767	1.501	[1.386, 1.853]	[0.855, 1.122]	improves	identical
rho1	32	-	setup_s	0.0295	0.0043	6.857	[2.49, 8.89]	[0.716, 1.334]	improves	identical
rho1	32	-	solve_s	0.0312	0.0053	5.836	[3.577, 7.177]	[0.708, 1.317]	improves	identical
rho1	36	-	setup_s	0.0417	0.0052	7.961	[5.937, 10.321]	[0.892, 1.334]	improves	identical
rho1	36	-	solve_s	0.0532	0.0142	3.745	[2.505, 3.912]	[0.698, 1.139]	improves	identical
rho1	40	-	setup_s	0.0521	0.0057	9.105	[7.272, 11.092]	[0.771, 1.489]	improves	identical
rho1	40	-	solve_s	0.1071	0.0588	1.822	[1.448, 2.075]	[0.675, 1.389]	improves	identical
rho1	44	-	setup_s	0.07	0.0064	10.959	[9.597, 13.659]	[0.964, 1.5]	improves	identical
rho1	44	-	solve_s	0.4518	0.3894	1.16	[0.979, 1.457]	[0.954, 1.206]	no measurable change	identical
rho1	48	-	setup_s	0.0908	0.007	13.023	[9.972, 18.749]	[0.811, 1.168]	improves	identical
rho1	48	-	solve_s	1.1729	1.0332	1.135	[1.07, 1.27]	[0.942, 1.045]	improves	identical
rho1	52	-	setup_s	0.1192	0.0074	16.094	[13.658, 25.896]	[0.897, 1.373]	improves	identical
rho1	52	-	solve_s	1.7896	1.648	1.086	[0.902, 1.151]	[0.83, 1.136]	no measurable change	identical
rho3	32	-	setup_s	0.0294	0.0043	6.774	[5.313, 9.146]	[0.715, 1.098]	improves	identical
rho3	32	-	solve_s	0.0283	0.0109	2.595	[1.978, 4.169]	[0.708, 1.359]	improves	identical
rho3	36	-	setup_s	0.083	0.0103	8.029	[6.499, 8.524]	[0.879, 1.121]	improves	identical
rho3	36	-	solve_s	0.0714	0.0364	1.961	[1.627, 2.097]	[0.799, 1.271]	improves	identical
rho3	40	-	setup_s	0.0968	0.0114	8.5	[6.877, 10.815]	[0.771, 1.135]	improves	identical
rho3	40	-	solve_s	0.0728	0.0304	2.393	[1.997, 3.793]	[0.661, 1.406]	improves	identical
rho3	44	-	setup_s	0.1393	0.0127	10.996	[9.052, 12.045]	[0.93, 1.313]	improves	identical
rho3	44	-	solve_s	0.1612	0.0928	1.737	[1.496, 2.166]	[0.981, 1.246]	improves	identical
rho3	48	-	setup_s	0.1872	0.0139	13.506	[11.184, 16.549]	[0.827, 1.317]	improves	identical
rho3	48	-	solve_s	0.2142	0.1233	1.737	[1.546, 2.195]	[0.947, 1.218]	improves	identical
rho3	52	-	setup_s	0.2186	0.0148	14.783	[11.676, 18.252]	[0.84, 1.113]	improves	identical
rho3	52	-	solve_s	1.2046	1.0852	1.11	[1.014, 1.144]	[0.909, 1.035]	no measurable change	identical

Samples and contention. Each row below gives the samples per configuration and the contended runs left out per arm (A, A′, B).

part	bits	kernel	samples	contended (A, A′, B)	pinned fields
bsgs1	40	chain	21	0, 0, 0	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	40	paired	21	1, 2, 2	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	44	chain	21	0, 1, 1	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	44	paired	21	0, 0, 1	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	48	chain	21	2, 1, 1	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	48	paired	21	0, 2, 0	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	52	chain	15	0, 0, 0	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs1	52	paired	15	0, 0, 0	answers_digest, table_entries, baby_steps, giant_steps, candidates, false_candidates
bsgs3	48	chain	21	1, 2, 0	answers_digest
bsgs3	48	paired	21	0, 1, 2	answers_digest
rho1	32	-	21	0, 0, 0	log, verified, total_steps, distinguished_points
rho1	36	-	21	0, 0, 1	log, verified, total_steps, distinguished_points
rho1	40	-	21	0, 0, 1	log, verified, total_steps, distinguished_points
rho1	44	-	21	0, 0, 0	log, verified, total_steps, distinguished_points
rho1	48	-	21	0, 0, 0	log, verified, total_steps, distinguished_points
rho1	52	-	15	0, 0, 0	log, verified, total_steps, distinguished_points
rho3	32	-	21	1, 0, 0	log, verified
rho3	36	-	21	1, 1, 0	log, verified
rho3	40	-	21	0, 0, 0	log, verified
rho3	44	-	21	0, 0, 0	log, verified
rho3	48	-	21	1, 0, 0	log, verified
rho3	52	-	15	0, 0, 0	log, verified

Field operations per visited point. `lit2026_w11_bsgs_opcount` produced 120 B rows, each compared with its A row. They are identical in multiplications, squarings, inversions, additions, points visited and slow rounds.

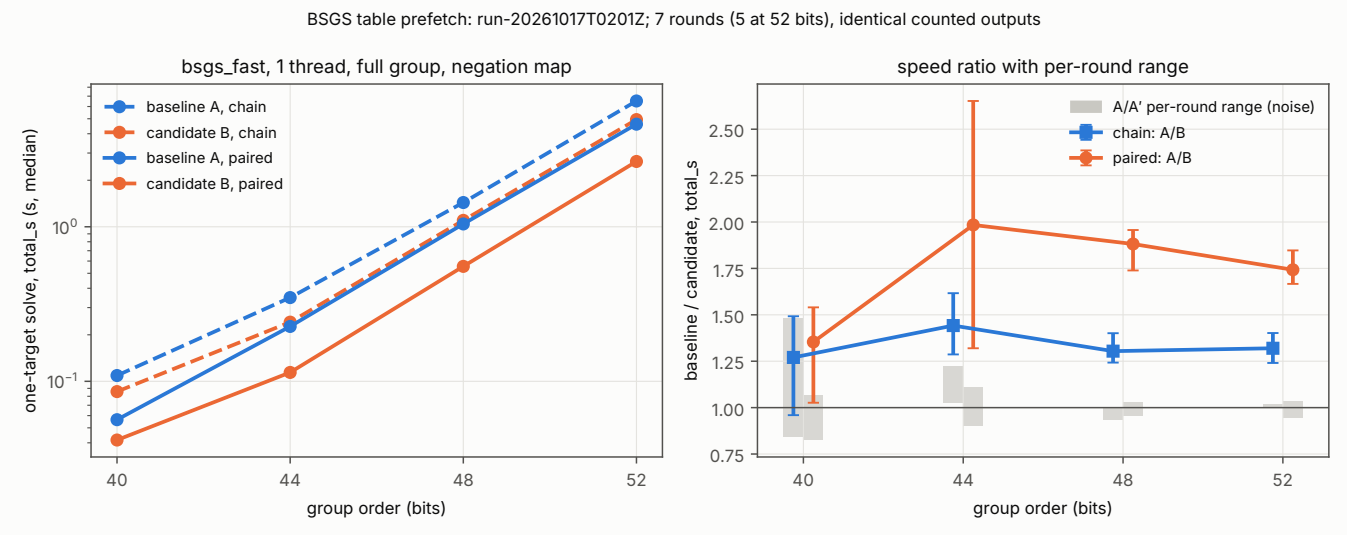


Figure 1: BSGS, run-20261017T0201Z: one-target `total_s` medians per arm (left) and A/B with the per-round range against the A/A′ range (right). 7 rounds, 5 at 52 bits.

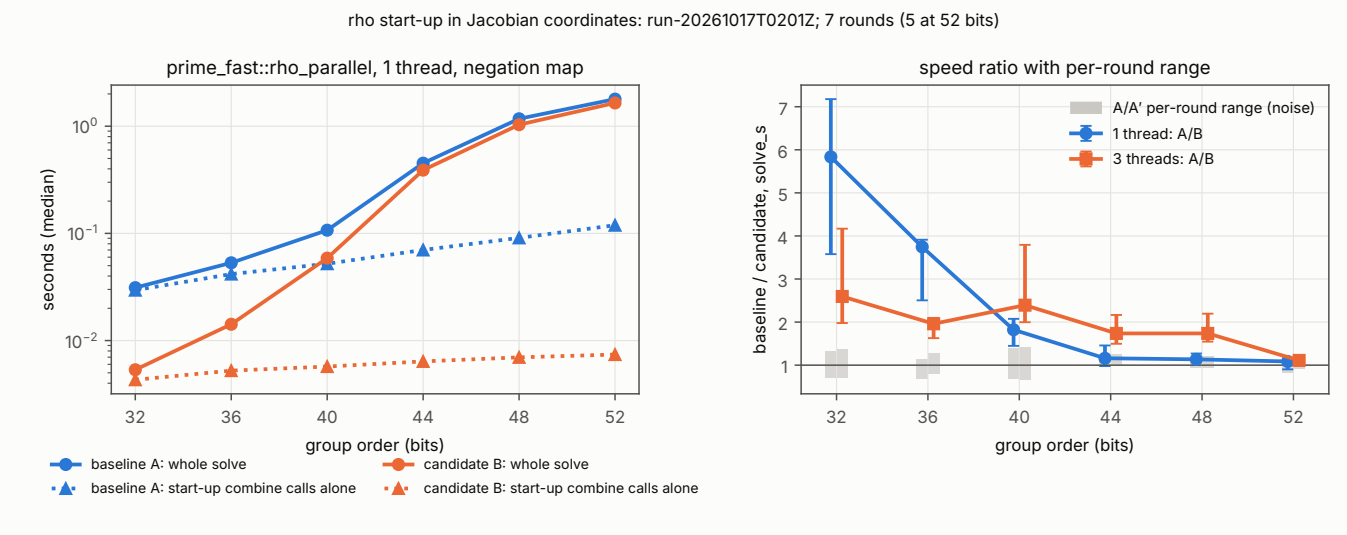


Figure 2: Rho, run-20261017T0201Z: `solve_s` and the start-up `combine` calls timed alone (left), and A/B with the per-round range at 1 and 3 threads (right).

1.3. External reference (PARI/GP `elllog`)

Same curves and targets, single-threaded. Times are medians, in seconds. PARI is timed by `getabstime()` around `elllog` alone.

curve role	bits	PARI med. (n)	A med.	B med.	B / PARI	A / PARI	PARI verified
bsgs	40	1.006 (3)	0.0565	0.0417	0.0415	0.0561	yes
bsgs	44	4.462 (3)	0.2265	0.1142	0.0256	0.0508	yes
bsgs	48	18.495 (3)	1.0429	0.5543	0.03	0.0564	yes
bsgs	52	27.749 (3)	4.6114	2.6462	0.0954	0.1662	yes
rho	32	0.032 (3)	0.0312	0.0053	0.1669	0.974	yes
rho	36	0.141 (3)	0.0532	0.0142	0.1007	0.377	yes
rho	40	0.679 (3)	0.1071	0.0588	0.0866	0.1577	yes
rho	44	3.162 (3)	0.4518	0.3894	0.1231	0.1429	yes
rho	48	11.754 (3)	1.1729	1.0332	0.0879	0.0998	yes
rho	52	80.788 (3)	1.7896	1.648	0.0204	0.0222	yes

The native arm is the paired kernel's `total_s` for the BSGS curves and `solve_s` for the rho curves, both at one thread.

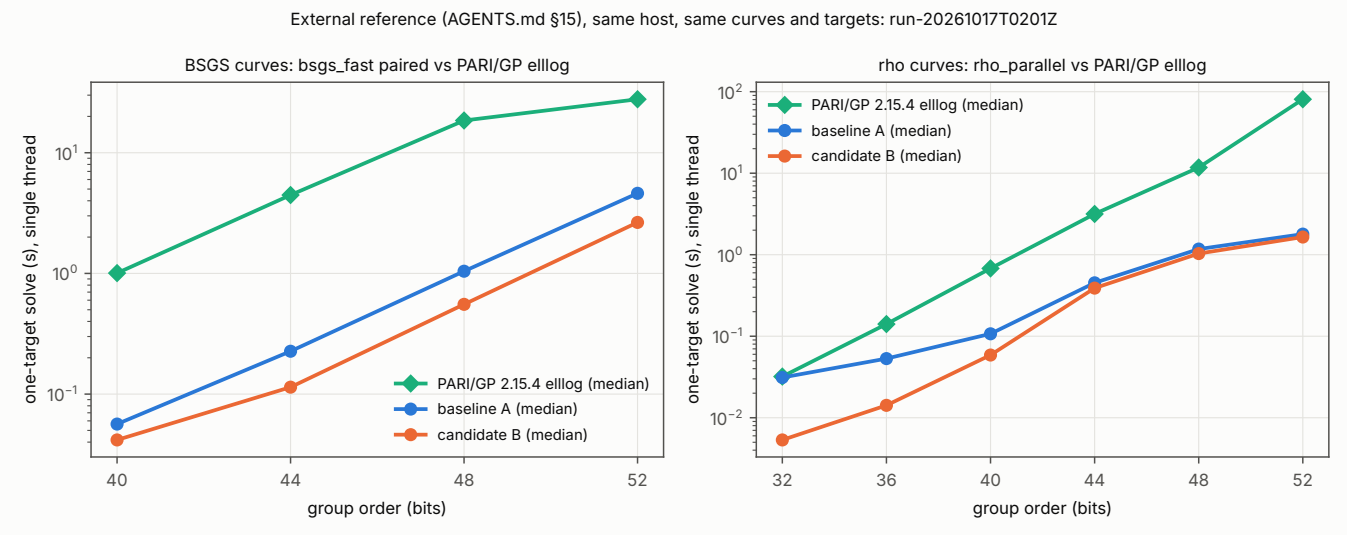


Figure 3: PARI/GP 2.15.4 `elllog` against both native arms on the same curves and targets, run-20261017T0201Z.

1.4. Raw records

`raw/bsgs1.jsonl`, `raw/bsgs3.jsonl`, `raw/rho1.jsonl`, `raw/rho3.jsonl`, `raw/pari.jsonl`, `raw/opcount.jsonl` and `raw/isolation.jsonl` (one isolation record per timed process, joined to its sample by label); `summary.json`; `manifest.json`; `interruptions.txt`.